Deep Learning (23AID211)

Image Denoising Using DL Models (Based on CNNs & Autoencoders)

By

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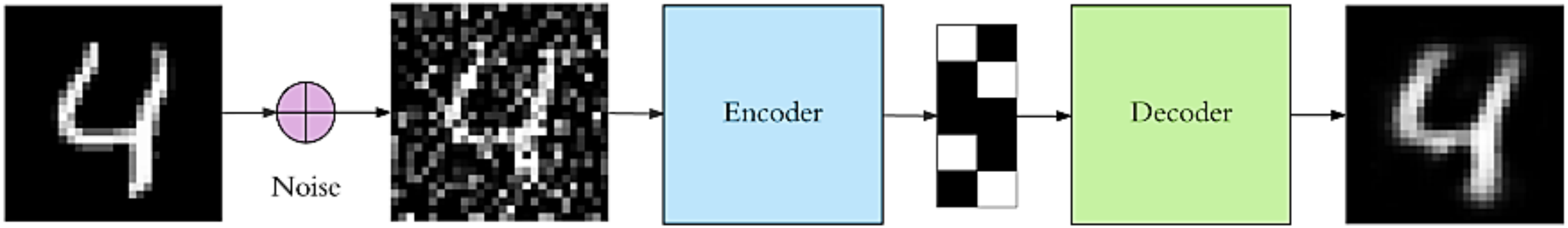
Faridabad Campus, Delhi NCR, India

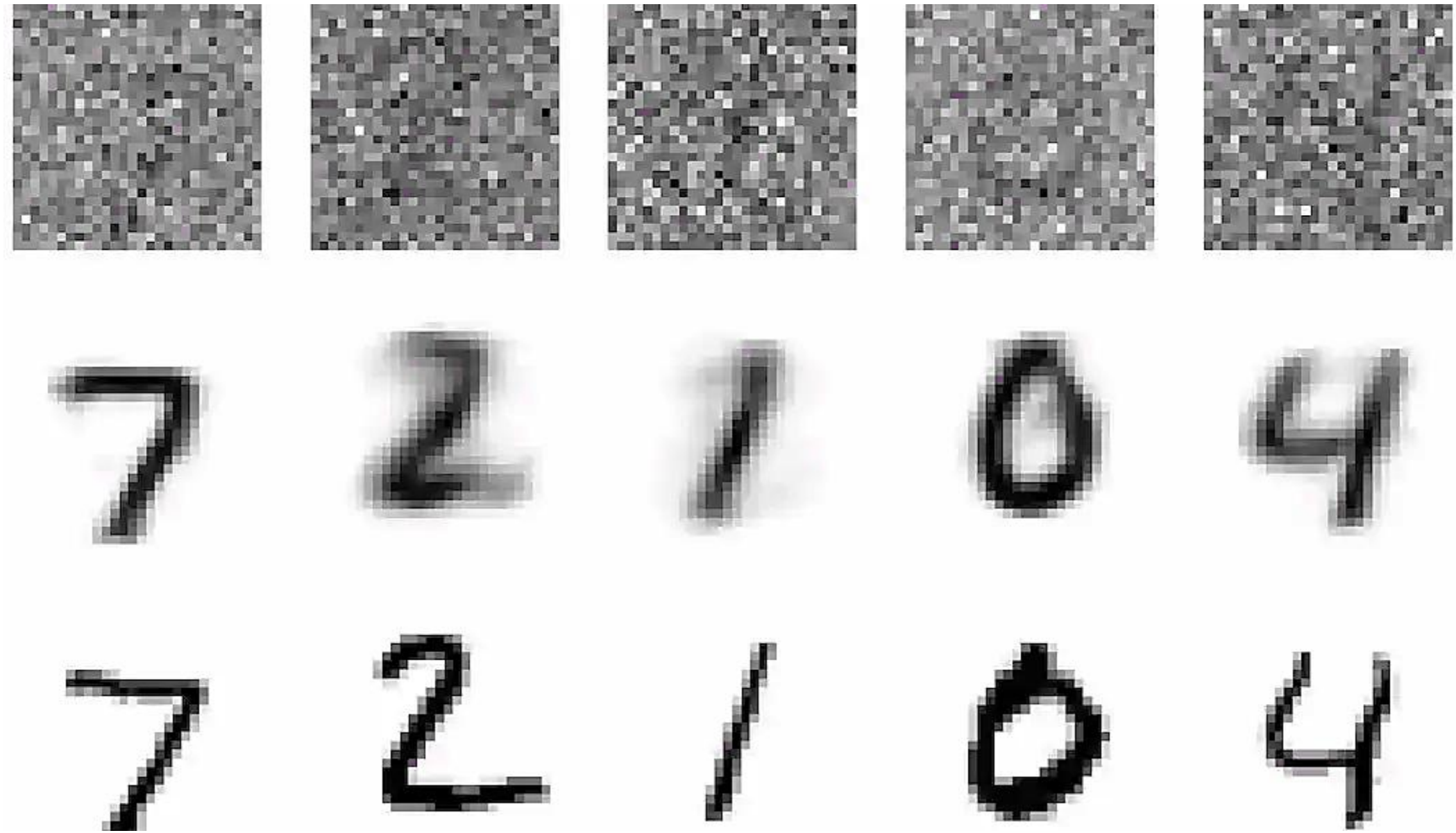
Bias and Variance in ML/DL Models Training

- A Denoising Autoencoder (DAE) is a type of neural network that removes noise from images by learning to map corrupted inputs to clean originals.



Noisy Sample





Various Terminology and Loss function

1. Fundamental Mathematical Formulation

A denoising autoencoder (DAE) consists of two main functions: the Encoder (E) and the Decoder (D).

- **Input (Noisy Image):** $x' = x + n$, where x is the clean image and n is the noise (e.g., Gaussian noise).
- **Encoder Mapping:** $h = E(x') = \sigma(Wx' + b)$, where h is the latent representation, W is the weight matrix, b is the bias vector, and σ is an activation function (like ReLU).
- **Decoder Mapping:** $\hat{x} = D(h) = \sigma'(W'h + b')$, where $\hat{x}$ is the denoised (reconstructed) image.
- **Objective:** The goal is to minimize the difference between the reconstructed image $\hat{x}$ and the original clean image x. GeeksforGeeks +4

Various Terminology and Loss function

2. Loss Function (Reconstruction Loss)

The autoencoder is trained by minimizing a loss function L, typically the **Mean Squared Error (MSE)**, between the network output $\hat{x}$ and the original clean image x: arXiv +4

$$L(x, \hat{x}) = ||x - \hat{x}||^2 = ||x - D(E(x + n))||^2$$

For a dataset of N images, the objective is to minimize the average loss:

$$J(W, b, W', b') = \frac{1}{N} \sum_{i=1}^{N} ||x_i - \hat{x}_i||^2$$

Various Terminology and Loss function

3. Convolutional Autoencoder (CAE) for Image Denoising

In image processing, convolutional layers are used instead of dense layers to preserve spatial information. Springer Nature Link +4

- **Convolutional Encoder:** $h_k = \sigma(x' * W_k + b_k)$, where k represents a feature map, and $*$ is the convolution operator.
- **Decoder/Deconvolution:** $\hat{x} = \sigma'(\sum_k h_k * W'_k + b'_k)$, often using transposed convolutions or upsampling layers. ResearchGate +3

Various Terminology and Loss function

4. Denoising Evaluation Metrics

- **Peak Signal-to-Noise Ratio (PSNR):** Measures the ratio between the maximum possible power of a signal and the power of corrupting noise, usually in dB.

$$\mathrm{PSNR} = 10 \cdot \log_{10} \left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}} \right)$$

- **Structural Similarity Index (SSIM):** Measures the similarity between two images (luminance, contrast, and structure), ranging from -1 to 1.

$$\mathrm{SSIM}(x, \hat{x}) = \frac{(2\mu_x\mu_{\hat{x}} + c_1)(2\sigma_{x\hat{x}} + c_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + c_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + c_2)}$$

Advantages

- Help models learn robust, meaningful features that are less sensitive to noise or missing data.
- Reduce the risk of merely copying input data (identity mapping), especially when compared to basic autoencoders.
- Improve performance on tasks such as image denoising, data imputation and anomaly detection by reconstructing clean signals from corrupted inputs.
- Enhance the generalizability of learned representations, making models more useful for downstream tasks.

Limitations

- May require careful tuning of the type and level of noise added to the inputs for optimal performance.
- Can be less effective if the noise model used during training does not match the type of corruption seen in real-world data.
- Like other unsupervised methods, provide no guarantees that learned features will be directly useful for specific downstream supervised tasks.

Non-Linear Filters

Linear filtering in computer vision modifies images by replacing each pixel value with a weighted sum of its neighbors, defined by a kernel or mask.

Characteristics and Types

Gaussian Filter: The Gaussian filter blurs the image, as the output is an average of the pixel values within the particular neighbourhood, with a weighting function of a Gaussian distribution.

Non-Superposition: These filters do not obey the superposition principle; the output for a sum of signals is not the sum of individual outputs.

Non-Linear Filters

Characteristics and Types

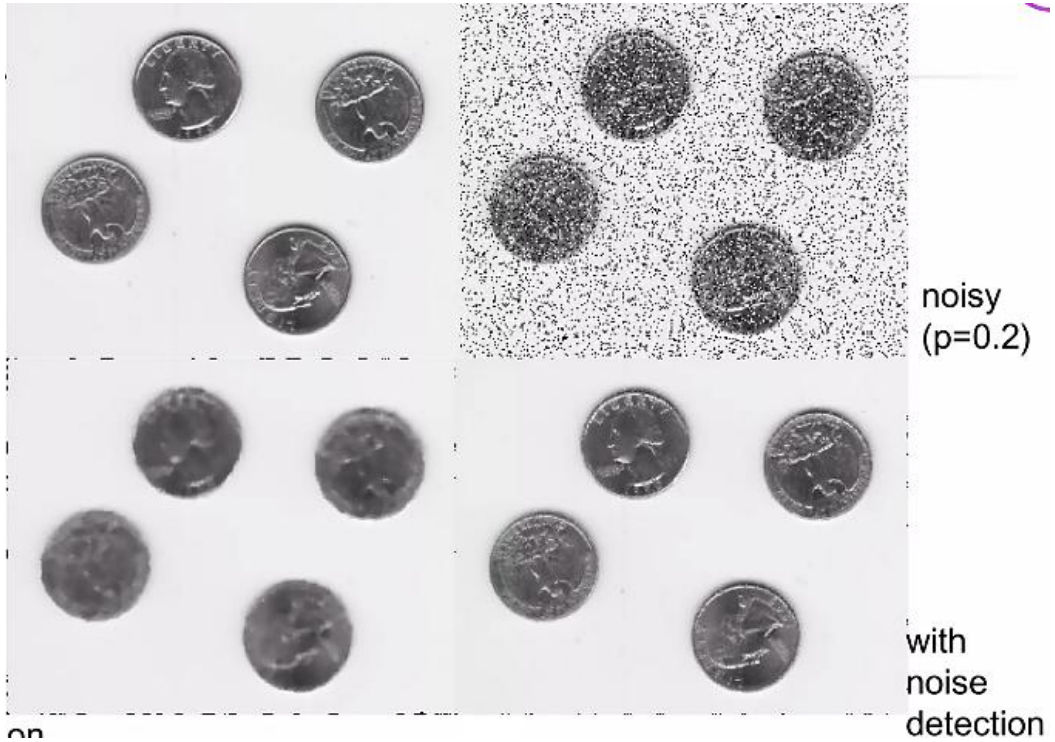
Median Filter: Selects the median value in a local neighbourhood, effective for removing "salt-and-pepper" noise without blurring edges.

Maximum/Minimum Filters: Select the highest or lowest value in a window, used to brighten/darken images or suppress positive/negative impulses.

Non-Linear Filters: (Mean Filtering)

225	225	225	226	226	226	226	226
225	225	255	226	226	226	225	226
226	226	225	226	0	226	226	255
255	226	225	0	226	226	226	226
225	255	0	225	226	226	226	255
255	225	224	226	226	0	225	226
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226	226	225	226	226	226	226	226

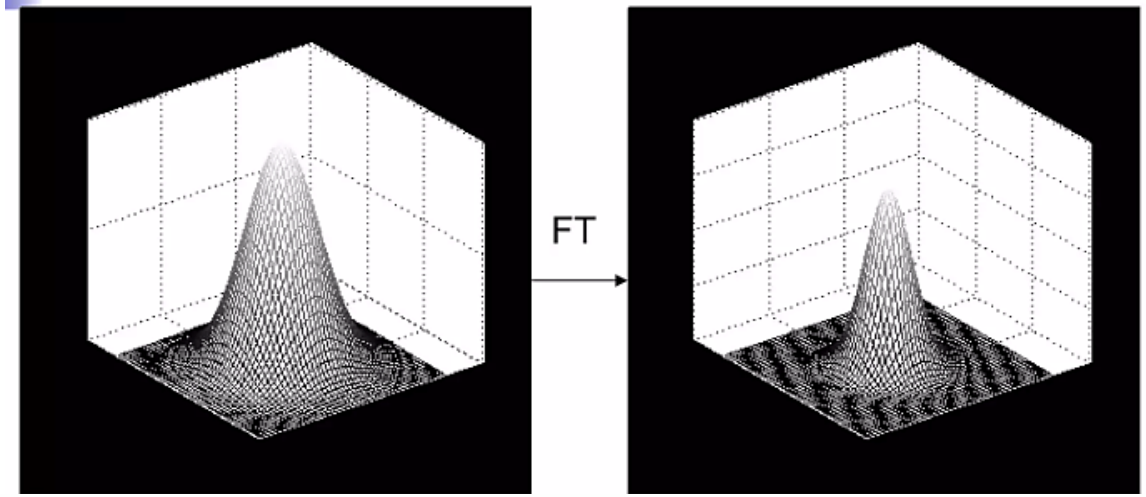
0	225	225	226	226	226	226	226
225	225	226	226	226	226	226	226
225	226	226	226	226	226	226	226
226	226	225	**225**	226	226	226	226
225	225	225	225	226	226	226	226
225	225	225	226	226	226	226	226
225	225	225	226	226	226	226	226
226	226	226	226	226	226	226	226



noisy
(p=0.2)

with
noise
detection

Non-Linear Filters: Gaussial Filtering



noisy



PSNR=20.2dB

(σ=25)

denoised



PSNR=24.4dB

(σ=1)

denoised



PSNR=22.8dB

(σ=1.5)

Matlab functions: imfilter, filter2

Linear and Non-Linear Filters

Non-linear filters are signal-processing algorithms in which the output is not a linear function of the input.

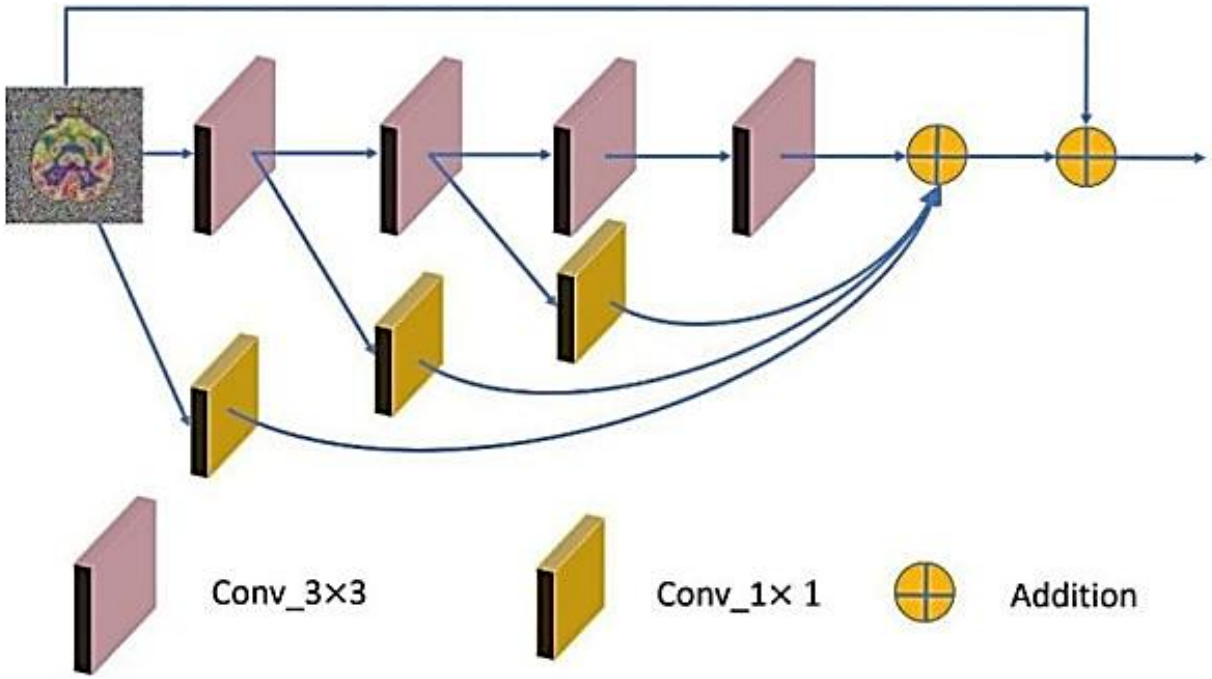
Characteristics and Types

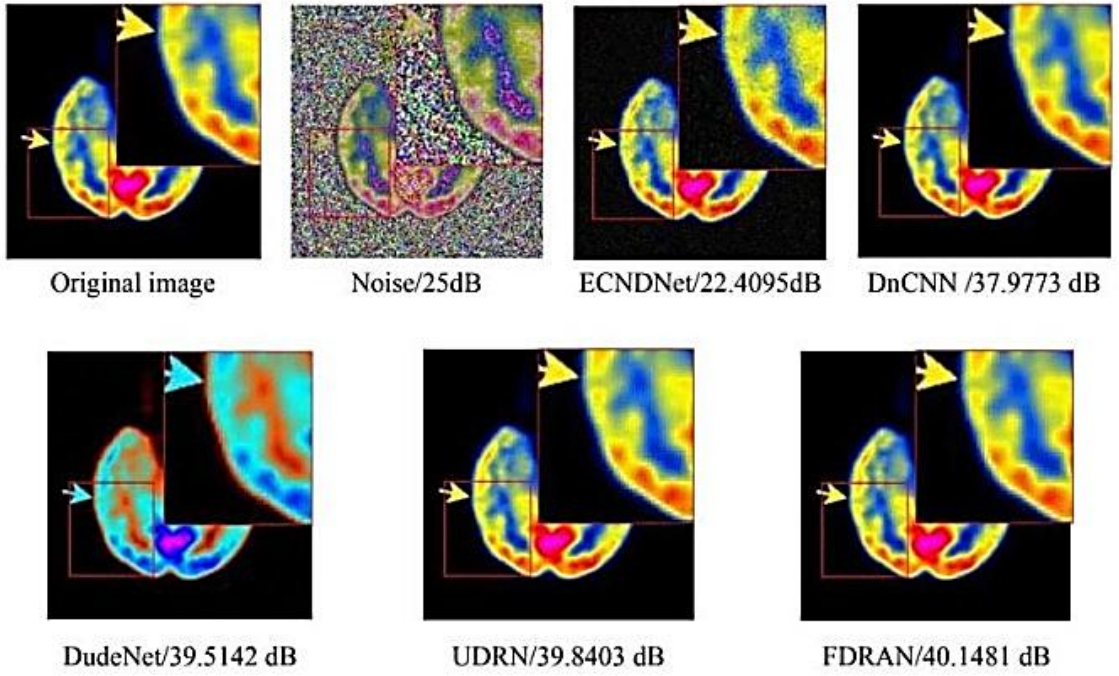
Convolution/Correlation: The kernel slides over the image. The output pixel is a weighted sum of the input pixels in the neighborhood.

- **Smoothing/Blurring:** Used to reduce noise or remove fine details.
- **Sharpening:** Enhances edges and fine details by emphasizing intensity differences.
- **Edge Detection:** Highlights significant intensity changes using operators like Sobel, Prewitt, or Laplacian filters.

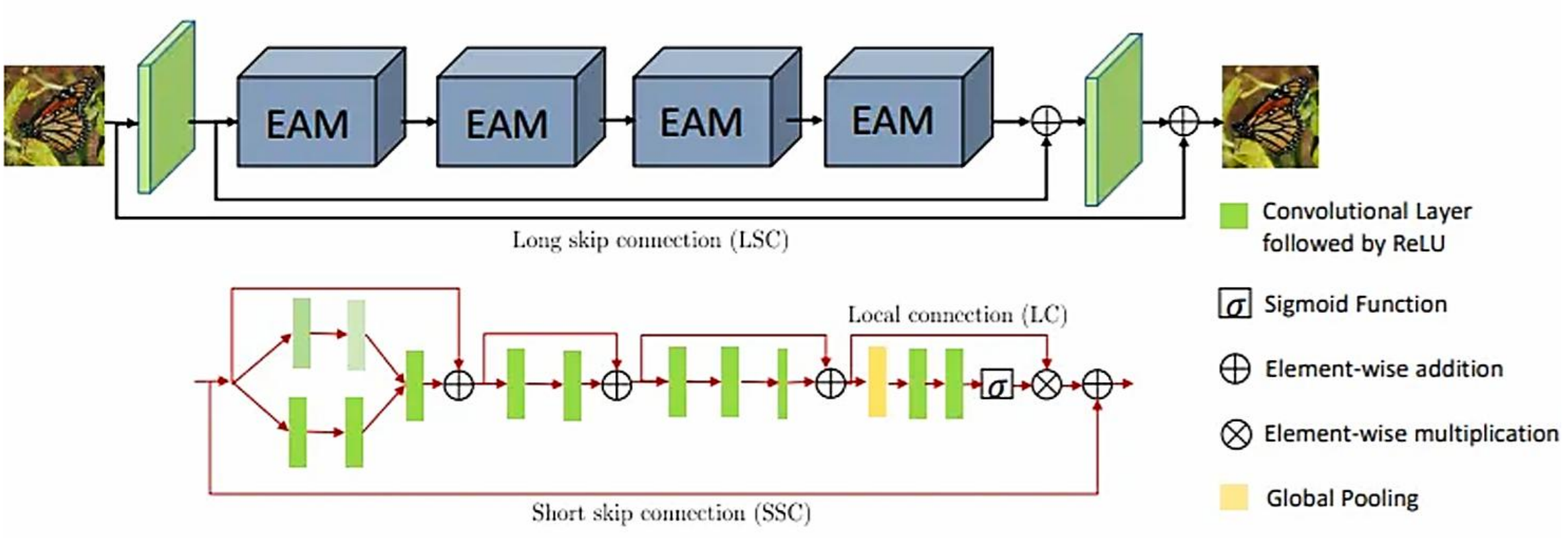
Advantages

- Noise in acquisition and transmission degrades image quality and affects the reliability of diagnosis and research.
- High-quality images are essential for disease detection, prognosis assessment, and treatment plan development.
- The overall network structure contains four parts: a global sparse network (GSN), a residual attention network (RAN), a feature distiller network (FDN), and a feature processing block (FPB).





Noise Removal Using Self-Attention



- In the realm of image denoising, the use of convolutional neural networks (CNNs) has lately gained traction.
- Low-light photography suffers from a decline in perceived image quality due to the limited dynamic range of pixel values.